7.9.7 Treatment and Control Measures

- Antibacterial therapy has been very effective in the treatment of patients.
- Ampicillin, amoxicillin and cotrimoxazole are useful in the treatment of typhoid fever.
- At present, ciprofloxacin is the drug of choice.
- Typhoid fever can be effectively controlled by sanitary measures for disposal of sewage, clean water supply and supervision of food processing and handling.

HOTS

Why is proper hand washing considered the most important element in controlling communicable infections?

7.10 *Vibrio Cholerae*

Vibrio is one of the curved rod bacteria, prominent in the Medical Bacteriology. They are present in marine environment and surface waters worldwide. *Vibrio* is a member of the family Vibrionaceae. The most important member of this genus is *Vibrio cholerae*, the causative agent of cholera. The term *Vibrio* is derived from **Vibrare** (Latin word) which means "**to shake or vibrate**" and the word **Cholera** is derived from **Chole** (Greek word) which means, "**to bile**" (Figure 7.16).

7.10.1 Morphology

Vibrio cholerae is gram negative, curved or comma shaped, (1.5um x 0.2–0.4um in size) non – capsulated. The organism is very actively motile with a single polar flagellum and the characteristic movement is called as **darting motility**. In stained smears of mucus flakes from acute cholera patients, the Vibriois seen arranged in parallel rows. This was described by Robert Koch as **"fish in stream"** appearance.

7.10.2 Culture Characteristics

Vibrio cholera is strongly aerobic. It grows best in alkaline media with the optimum temperature 37°C and pH 8.2. It is non-halophilic, therefore, cannot grow in media with a concentration of sodium chloride more than 7% (Figure 7.17). Some of the media in which *Vibrio cholerae* are cultivated are tabulated below in Table 7.16.

Table 7.16: Colony morphology of Vibrio *cholerae* on various media

Media	Colony morphology
Nutrient agar	The colonies are moist, translucent round disks (1–2mm in diameter) with a bluish tinge in transmitted light.
MacConkey agar	The colonies are colorless at first but become reddish on prolonged incubation due to late fermentation of lactose.
Thiosulphate citrate bile sucrose agar (pH 8.6)	It is used as a selective medium for isolation of *Vibrios*. It produces large yellow convex colonies due to sucrose fermentation.

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